

Scientific Notation

Scientific Notation is used to express very large numbers, or very small numbers, in a compact form.

Expression	Common Decimal Notation	Scientific Notation
124.5 million kilometres = 124.5 billion meters	124 500 000 km = 124 500 000 000 m	1.245×10^8 km = 1.245×10^{11} m
154 thousand nanometres	154000 nm = 0.000 154 m	1.54×10^5 nm = 1.54×10^{-4} m

1 meter = 1 000 000 000 nanometers

$$1.54 \times 10^{-4} \text{ m}$$

Ex) Write the following values in scientific notation:

- a) 35 000 3.5×10^4
- b) 0.00000492 4.92×10^{-6}
- c) 35 3.5×10
- d) 1240 1.24×10^3